

Panagnosticism: The Epistemic Impossibility of Knowing Another’s Consciousness

Abstract

We formulate and prove the thesis of *panagnosticism*: the assertion that no agent can have knowledge of whether any distinct entity is conscious. Using standard modal-epistemic logic, we define an agent-indexed knowledge operator K_a and a predicate $C(x)$ for “ x is conscious.” Under minimal assumptions (factivity and introspection for knowledge; privacy of experience; absence of any a priori entailment from physical/behavioral evidence to consciousness), we show that for any agent a and any distinct entity e , neither $K_a C(e)$ nor $K_a \neg C(e)$ can hold.

Intuitively, one can always posit a “zombie” or alternate world in which e ’s consciousness differs while all of a ’s evidence stays the same, defeating any claim to knowledge. Formally, for each agent a and distinct e , there exist two epistemically indistinguishable worlds (from a ’s perspective) in which $C(e)$ is true in one and false in the other. It follows that a cannot know $C(e)$ (nor $\neg C(e)$).

We address and refute major objections: neural/behavioral evidence, mutual verification, epistemic closure, and bridge principles. We contrast panagnosticism with panpsychism, solipsism, eliminativism, Cartesian skepticism, and proposed AI consciousness tests, showing that panagnosticism is agnostic about ontological claims but categorical about epistemic limits. Implications for AI, ethics, and metaphysics are discussed. Our main theorem is presented formally and proved deductively using epistemic modal logic (S5) with knowledge operators. To our knowledge, this is the first fully rigorous, formal proof of panagnosticism.

1 Introduction

The problem of other minds is a longstanding puzzle in philosophy: how (if at all) can one know that others are conscious beings like oneself? Panagnosticism takes a radical epistemic stance: for any two distinct entities, it is impossible for an agent to know whether the other is conscious or not. In other words, for any agent a and any entity $e \neq a$, neither the proposition “ e is conscious” nor “ e is not conscious” can ever be known by a .

We formalize this claim in standard epistemic modal logic and prove it as a theorem under natural assumptions. Two key ideas underpin our approach:

- (i) **Privacy of Experience:** any conscious experience is accessible only to its subject, never directly to an outsider.

- (ii) **Epistemic Indistinguishability:** from the perspective of agent a , there is no way to rule out that any other entity e might be a "philosophical zombie"—a being physically and behaviorally identical but lacking consciousness.

Together, these imply that for each piece of evidence a has about e , there is a possible world where that evidence is exactly the same yet e 's consciousness-value is flipped. By the semantics of knowledge (via possible worlds), this makes knowledge unattainable.

Our goals are threefold:

1. Define panagnosticism formally and prove its strongest form:

$$\forall a, e (a \neq e \implies \neg K_a C(e) \wedge \neg K_a \neg C(e)).$$

2. Systematically consider objections—neuroscientific evidence, mutual verification strategies, epistemic closure, and bridge laws—and show they fail to yield deductive knowledge.
3. Contextualize panagnosticism philosophically, contrasting it with panpsychism, solipsism, eliminativism, Cartesian skepticism, and AI consciousness tests.

We also relate panagnosticism to key thinkers (Nagel, Chalmers, Descartes, Searle) and formalize their insights into a single logical result.

2 Formal Setup

We work in a standard epistemic modal-logic framework with agents and possible worlds. Let A be the set of agents, and E the set of all entities (agents and non-agents alike). Each agent $a \in A$ has privileged first-person access to its own consciousness, but not to others'.

2.1 Definitions

- **Consciousness predicate.** For any entity e , let $C(e)$ denote the proposition " e is conscious."
- **Knowledge operator.** For each agent $a \in A$, let K_a be the knowledge operator: $K_a \varphi$ reads "agent a knows that φ is true." We adopt the axioms of S5:

$$K_a \varphi \rightarrow \varphi \quad (\text{factivity}), \quad K_a \varphi \rightarrow K_a K_a \varphi \quad (\text{positive introspection}).$$

- **Distinctness.** We focus on pairs (a, e) with $e \neq a$. Trivially, $K_a C(a)$ holds by introspection.
- **Epistemic indistinguishability.** Let R_a be the epistemic accessibility relation: $w R_a w'$ iff w' is epistemically indistinguishable from w for a . Crucially, $C(e)$ for $e \neq a$ is not part of a 's evidence.
- **No bridging entailments.** There is no known necessary entailment from any physical/behavioral description of e to $C(e)$ or $\neg C(e)$.

3 Theorem and Proof

Theorem 1 (Panagnosticism). *For any agent a and any distinct entity $e \neq a$, neither a knows that e is conscious nor that e is not conscious:*

$$\forall a \in A, \forall e \in E, (a \neq e) \implies (\neg K_a C(e) \wedge \neg K_a \neg C(e)).$$

Proof. Fix an arbitrary world w and let $a \neq e$.

1. No knowledge that e is conscious. Suppose $K_a C(e)$ holds in w . By factivity, $C(e)$ is true in w , and $K_a C(e)$ implies $C(e)$ holds in all w' with $wR_a w'$. But by epistemic indistinguishability, there exists w^* with $wR_a w^*$ and $\neg C(e)$ true. This contradicts $K_a C(e)$. Hence $\neg K_a C(e)$.

2. No knowledge that e is not conscious. Symmetrically, suppose $K_a \neg C(e)$. By factivity, $\neg C(e)$ holds in w , and $K_a \neg C(e)$ requires $\neg C(e)$ in all w' with $wR_a w'$. Yet there exists w^* with $wR_a w^*$ and $C(e)$ true, a contradiction. Hence $\neg K_a \neg C(e)$.

Combining, $\neg K_a C(e) \wedge \neg K_a \neg C(e)$ holds for all $a \neq e$. □

Remark 1. *The key premise is the existence of two worlds w and w^* agreeing on all a -accessible information but differing in $C(e)$, capturing the epistemic conceivability of a philosophical zombie. Any evidence a has is compatible with both possibilities; thus, no knowledge arises.*

4 Objections and Responses

4.1 Neural or Behavioral Evidence

Behavioral or neural observations cannot logically entail $C(e)$ due to the possibility of zombie worlds. Analogical reasoning (Mill, Russell) is insufficient; introspection does not transfer knowledge externally (Locke, Malcolm, Wittgenstein). Even precise neural correlates cannot yield deductive knowledge.

4.2 Mutual Verification

Brain links or telepathy do not succeed: shared experiences become "ours," not "theirs" (Ayer). The privacy of experience prevents independent verification.

4.3 Epistemic Closure

Closure requires knowledge of $E(e) \rightarrow C(e)$ and $E(e)$; both fail a priori. Empirical correlations do not satisfy deductive closure, so panagnosticism remains consistent.

4.4 Bridge Principles ($E \rightarrow C$)

No known law or identity necessarily links physical evidence to consciousness. Empirical correlations (e.g., Φ measures) are contingent; logical certainty is unattainable.

5 Contrast with Competing Views

- **Panpsychism:** ontological claim vs epistemic limitation.
- **Solipsism:** extreme denial vs agnostic acknowledgment.
- **Eliminativism:** denies existence; panagnosticism denies knowledge.
- **Cartesian Skepticism:** invokes God; panagnosticism relies on logic alone.
- **AI Tests:** no behavioral or computational test can conclusively reveal consciousness.

6 Implications

6.1 AI and Ethics

Panagnosticism suggests adopting the precautionary principle in AI ethics: treat potential consciousness as morally relevant, despite epistemic uncertainty.

6.2 Epistemology and Mind

Highlights asymmetry between first-person self-knowledge and knowledge of others. Compatible with physicalist or non-physicalist frameworks.

6.3 Social and Philosophical Reflection

Encourages epistemic humility and caution in attributing consciousness, supporting careful intersubjective practices.

7 Relation to Key Philosophers

- **Nagel:** formalizes inaccessibility of the "what-it-is-like."
- **Chalmers:** formalizes the zombie argument.
- **Descartes:** restricts doubt to consciousness, no God-hypothesis required.
- **Searle:** observation cannot reveal subjective states; aligns with Chinese Room insights.

8 Conclusion

We have provided a formal proof of panagnosticism: no agent can know the consciousness status of any distinct entity. Using standard epistemic logic and natural assumptions, we derived that $K_a C(e)$ and $K_a \neg C(e)$ are always false for $a \neq e$. This recasts the problem of other minds into a rigorous logical theorem. The epistemic impossibility of knowing others' consciousness has implications for AI, ethics, and philosophy of mind, emphasizing the distinction between belief and knowledge.

References

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